Introduction to Artificial Intelligence

COSC3030

Course Projects

Deliverables

1- A full report.
2- A source code
3- A demonstration

You report must contain all the aspect of your software development, emphasizing in particular the justification, the description, and the use of a particular AI technique.

Marking Scheme

Report	40%
Code	40%
Demo	20%

Deadline

Wedensday 12 November EOB

Submit your material in Bb

Notes:

This is a group project (4 students maximum)

All the projects must be selected.

Project 1: The Nim game

Objectives:

The goal of this assignment is to deepen your understanding of search techniques, and to put them into a practice via their integration into a simple software game, namely, the Nim game.

The Nim game

The Nim game is a two-player game, where each player takes turns removing objects from one or more piles. The player requires to take the last object loses the game.

Outcomes

After this assignment, the student should be able to:

1. Design and implement a tree structure that encodes the search space of a given search problem.
2. Implement and use a standard tree search techniques.
3. Demonstrate how to use search techniques concepts to build a winning strategy.
4. Describe the advantages and the limitations of a search technique.

Problem

A game software company approached you asking to design for them a software for the Nim game. The requirement as follows

1. The game is to be played by the user versus the machine
2. It should be interactive
3. The objects (or their symbols) should be displayed
4. The software should give to the user the option whether or not he wants to make the first move.

Also, as the software is meant to be used by kids, the game should look simple, and not taking so too much time. For this purpose, the number of objects should be set to 15, and each player can take only one, two, or three objects from the pile.

Tasks

Write a program that implements a solution o the above problem. **Your solution must use a search technique of your choice.**

Marking criteria:

platform: by order of preference, mobile app, web app then desktop app

The number of piles : the larger the best.

Project 2: The Tic-Tac-Toe game

Objectives:

The goal of this assignment is to deepen your understanding of search techniques, and to put them into a practice via their integration into a simple software game, namely, Tic-Tac-Toe.

The Tic-Tac-Toe game

Tic-tac-toe is a game for two players who take turn marking the spaces in a 3×3 grid, each with his/her own mark. The player who succeeds in placing three of their marks in a horizontal, vertical, or diagonal row wins the game. The following example game is won by the first player using the mark X

Outcomes

After this assignment the student should be able to:

5. Design and implement a tree structure that encodes the search space of a given search problem.
6. Implement and use a standard tree search techniques.
7. Demonstrate how to use search techniques concepts to build a winning strategy.
8. Describe the advantages and the limitations of a search technique.

Problem

A game software company approached you asking to design for them a software for the Tic-Tac-Toe game. The requirements are as follows

5. The game is to be played by the user versus the machine
6. It should be interactive
7. The objects (or their symbols) should be displayed
8. The software should give to the user the option whether or not he wants to make the first move.

Also, as the software is meant to be used by kids, the game should look simple, and not taking too much time.

Tasks

Write a program that implements a solution o the above problem. **Your solution must use a search technique of your choice.**

Marking criteria:
platform: by order of preference, mobile app, web app then desktop app
The number of piles : the larger the best.

Project 3: Vision-based class attendance

Objectives:

The project focuses on designing an automated class attendance system utilizing existing CCTV cameras. This system aims to streamline the attendance-taking process by leveraging the surveillance infrastructure already present in classrooms or at the classroom entrance**.**

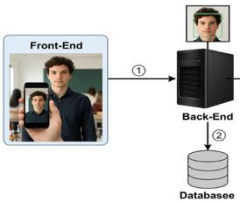
Outcomes

A software that can be used to check student attendance.

General pipeline

The system should work in an authentication-like manner**:**

1- The system detects faces
2- Extract a face signature using the deep network developed in [1]
3- Compare the signature with each face's signature to the gallery. The gallery is the dataset containing the students enrolled in the class. The comparison can be expressed using the cosine distance metric.

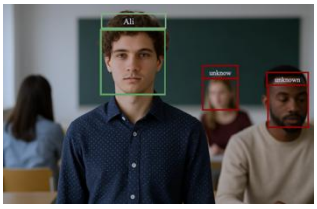


Enrollment mobile application

The enrollment (registration), should be performed as follows
The face of the person is taken by a mobile camera, with ID number, cropped then sent to the databases on the server.

System validation

The system will be validated as follows: Given a set of images or video containing multiple individuals, some enrolled in the system and others not, the authentication system should correctly detect and recognize the enrolled subjects, while assigning the label "unknown" to those who are not enrolled. The system must also be able to identify individuals under varying conditions, such as different distances from the camera, diverse lighting environments, and varying face poses.



Performance Evaluation & Comparison

Evaluate your model in terms of: Accuracy, Precision/Recall, F1-score and any other relevant metrics.

Resources:

Mobile phone

CCTV camera for the validation

The benchmark code (FaceNet) for our face recognition can be found in GitHub link https://github.com/CVML-KU-Research/FaceNet.

Project 4: Predicting Diabetes Using Artificial Neural Networks

Objectives:
Diabetes is a chronic disease with serious health complications if left unmanaged. Early detection can significantly improve patient outcomes. This project aims to build an Artificial Neural Network (ANN) model to predict diabetes based on patient health records. Use the Pima Indians Diabetes Dataset (UCI Repository / Kaggle)

Outcomes
A software (GUI) using an ANN to predict Diabetes

Tasks
You will use the widely used Pima Indians Diabetes Dataset, which contains diagnostic measurements and diabetes outcomes. You will explore dataset preprocessing, ANN implementation, and model evaluation**. You can follow these steps**

- Dataset Familiarization & Preprocessing
- ANN Design & Evaluation (1 or 2 hidden layers)
- Performance Evaluation & Comparison
  Evaluate your model in terms of: Accuracy, Precision/Recall, F1-score

Project 5: Predicting Diabetes Using Decision Tree

Objectives:
Diabetes is a chronic disease with serious health complications if left unmanaged. Early detection can significantly improve patient outcomes. This project aims to build a Decision Tree (DT) model to predict diabetes based on patient health records. Use the Pima Indians Diabetes Dataset (UCI Repository / Kaggle)

Outcomes
A software (GUI) using a Decision Tree to predict Diabetes

Tasks
You will use the widely used Pima Indians Diabetes Dataset, which contains diagnostic measurements and diabetes outcomes. You will explore dataset preprocessing, DT implementation, and model evaluation**. You can follow these steps**

- Dataset Familiarization & Preprocessing
- DT Design & Evaluation (1 or 2 hidden layers)
- Performance Evaluation & Comparison
  Evaluate your model in terms of: Accuracy, Precision/Recall, F1-score